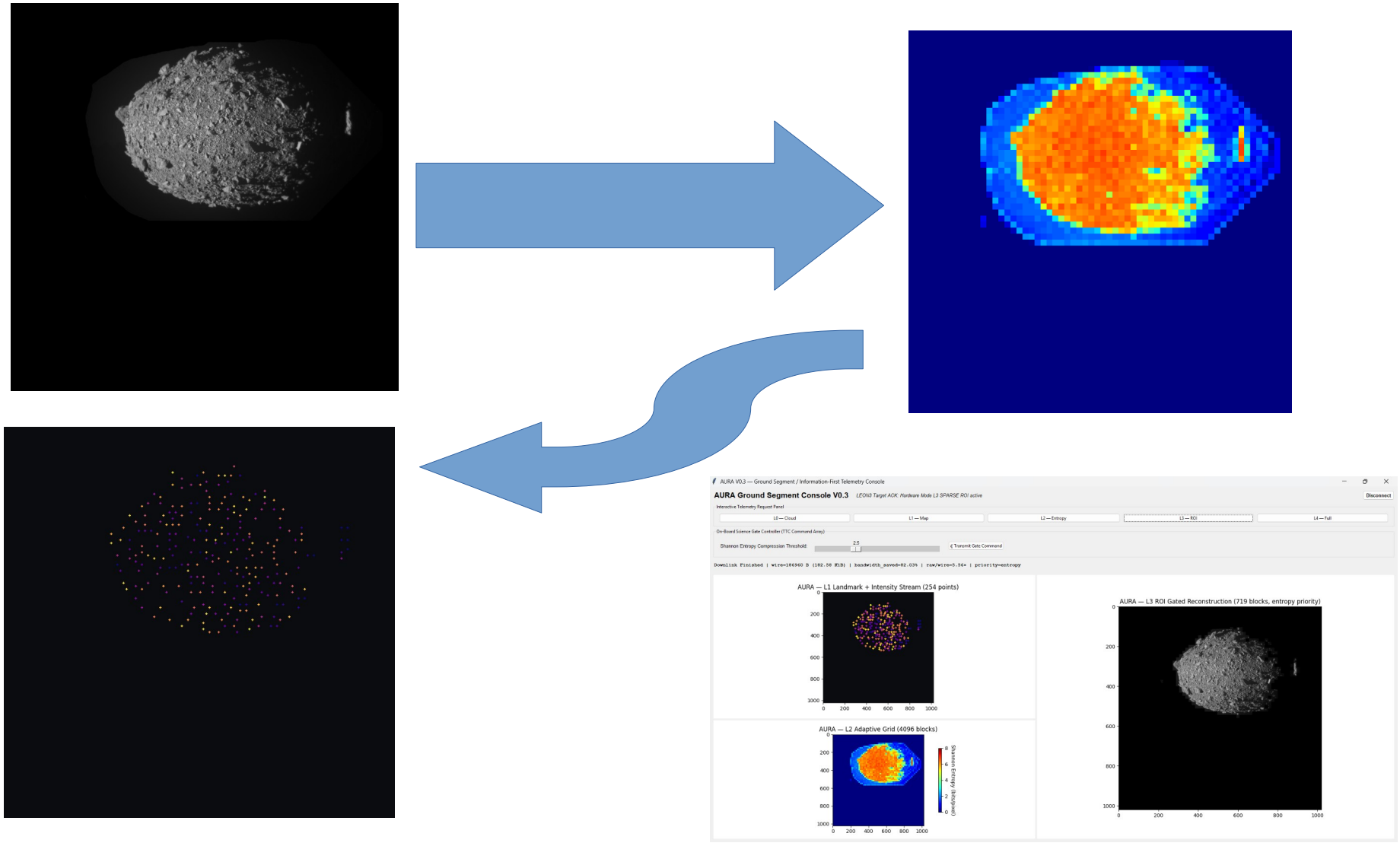
AURA: Autonomous Unsupervised Feature-Tracking for Real-Time Deep-Space Navigation



The image of the sky body was provided as part of the DART's Didymos Reconnaissance*(NASA).*

Who Am I?

Serhii Trush Embedded Systems Engineer | Electronics Developer

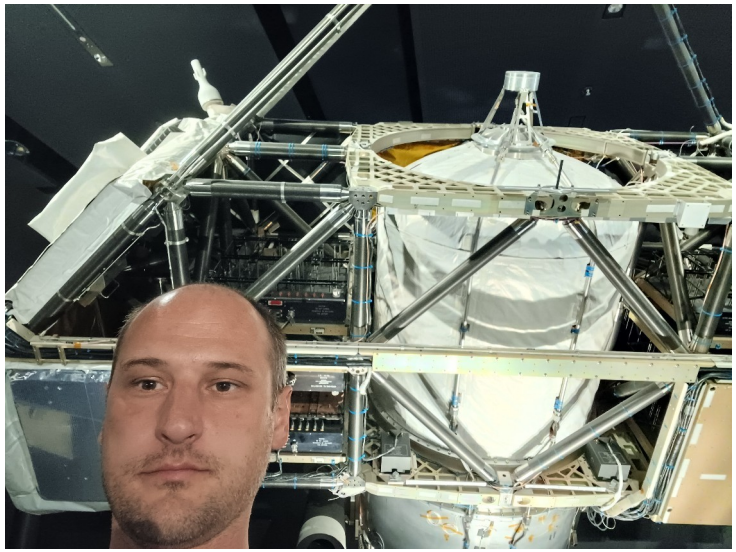
- GitHub: https://github.com/techn0man1ac/
- LinkedIn: https://www.linkedin.com/in/techn0man1ac/
- YouTube: https://www.youtube.com/@Techn0man1ac

Education

- Computer Systems Engineer(**Bachelor degree**)
- Engineering background(Commercial / Industrial electronics)
- Electronics and embedded systems
- Software development

Experience

- **10+ years in electronics**
- Embedded development
- Hardware design
- Software engineering



I'm standing next to the CRISTA-SPAS at the Deutsches Museum.

Origin of the Name “AURA”

The name was selected during an AI-assisted brainstorming process.

It resonated because it reflects the central idea of the project:

- Information awareness
- Detection of hidden structures
- Environmental perception
- Prioritization of meaningful information



Removing image blocks with low information

Project Connection

AURA attempts to identify and prioritize meaningful information inside large sensor datasets.

Instead of treating all pixels equally, the system focuses on regions carrying the highest informational value.

Interpretation

The term *AURA* can be viewed as the informational footprint of an observed environment.

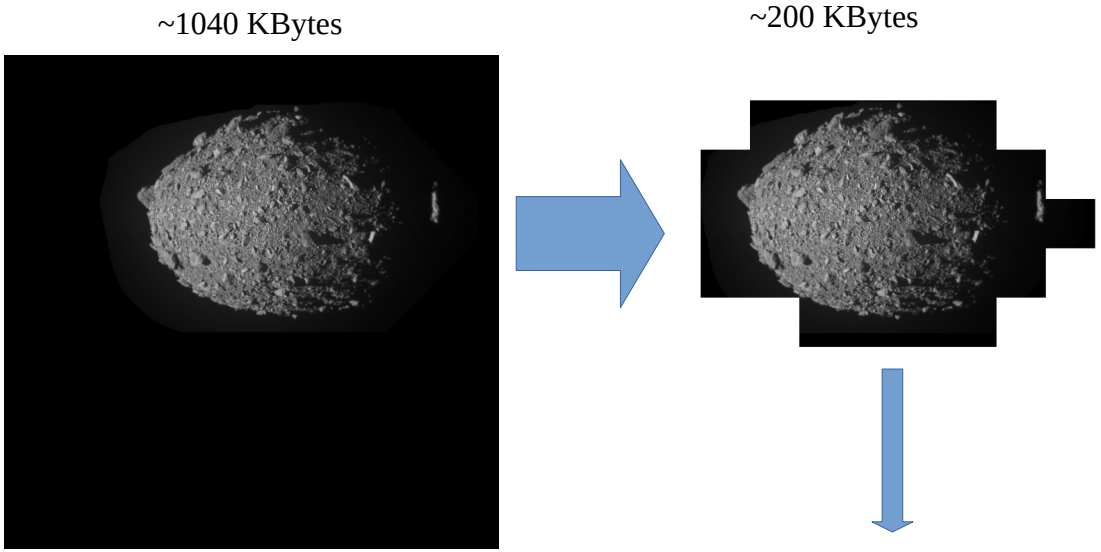
Connection to the Project

AURA attempts to identify and prioritize meaningful information inside large sensor datasets.

The Communication Problem

Deep Space Constraints

- Long communication delays
- Limited bandwidth
- Limited onboard computing resources
- Power constraints



Example

Transmitting a full image may require significantly more resources than transmitting a compact information summary.

Research Question

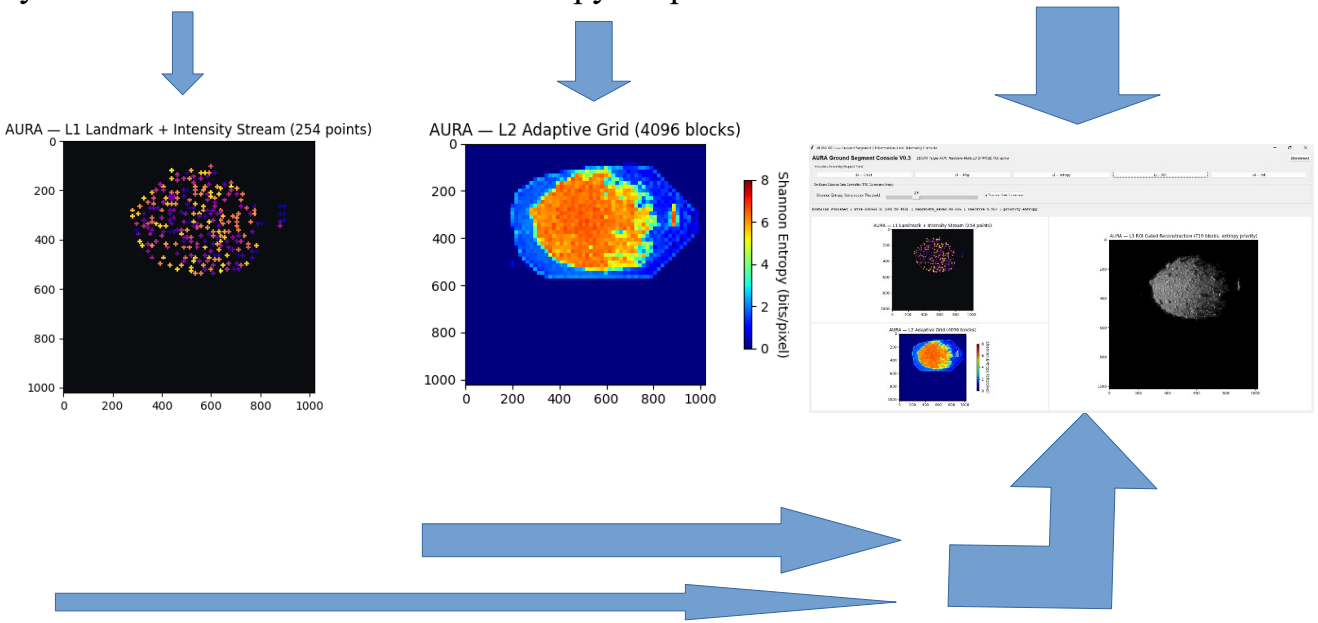
Can a spacecraft decide what information is worth transmitting before sending the full image?

Core Idea

Can pixels control everything? Can entropy take responsibility?
Not all pixels carry the same amount of information.

AURA Workflow

Cameras image → Information Analysis → Feature Point Cloud → Entropy Map → Decision → Data Transmission

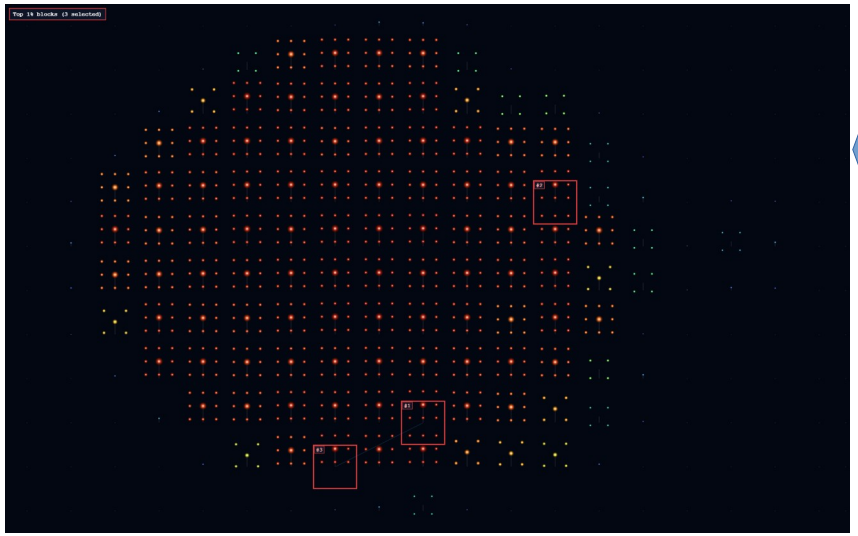


AURA goal:

- Reduce unnecessary data transmission while preserving scientific value.

Next is space navigation?

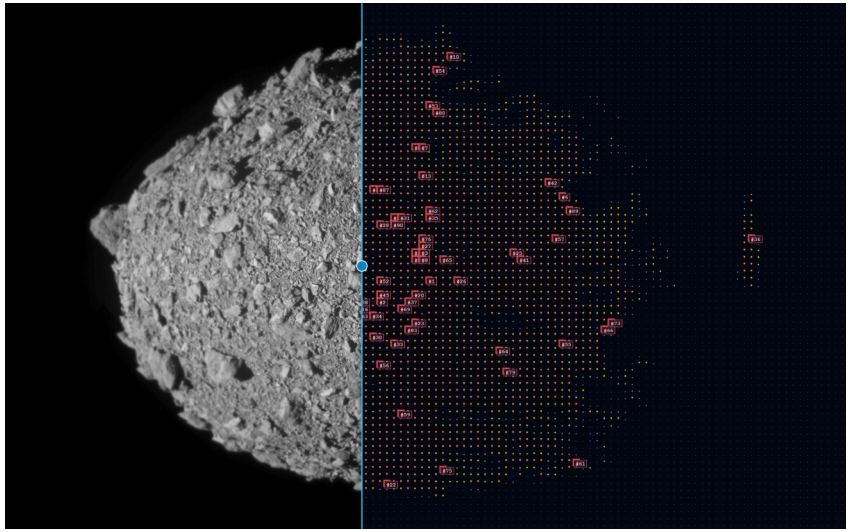




```
array([[0.32, 0.21, 0.05, ..., 0.05, 0.34, 0.39],
       [0.21, 0.08, 0.03, ..., 0.  , 0.13, 0.31],
       [0.15, 0.  , 0.  , ..., 0.  , 0.  , 0.23],
       ...,
       [0.  , 0.  , 0.  , ..., 0.  , 0.  , 0.  ],
       [0.  , 0.  , 0.  , ..., 0.  , 0.  , 0.  ],
       [0.  , 0.  , 0.  , ..., 0.  , 0.  , 0.  ]], shape=(68, 68))
```

Can entropy-derived representations support spacecraft perception and decision making?

Google AI Studio Sandbox



Purpose?	Why It Matters?
Fast experimentation environment for: • Entropy visualization • Feature extraction • Point cloud generation • Constellation generation • Concept demonstration	Allows rapid iteration before implementing ideas on embedded hardware. • Real-time entropy mapping • Point cloud(Constellation visualization) • Quick idea validation

Technical Details

Image Processing

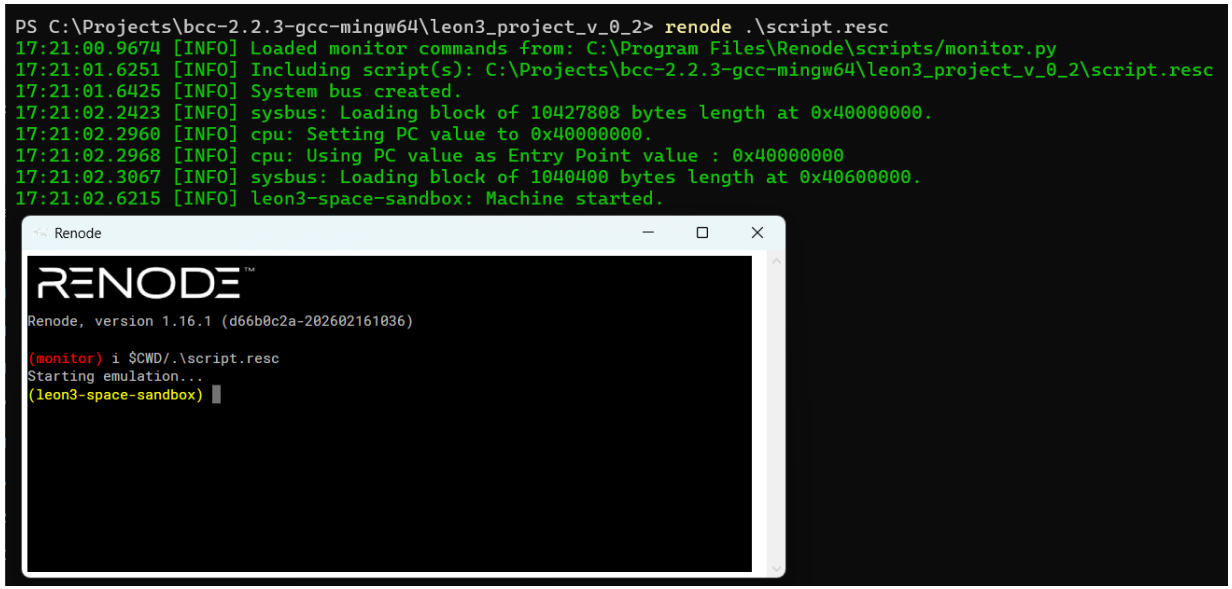
- Local entropy calculation
- Contrast maxima detection
- Sparse feature extraction

Embedded Optimizations

- Bit-wise operations
- Memory-aware design
- LEON3 compatibility
- Low computational overhead

Current Status

- Emulator implementation
- Functional prototype
- Experimental visualizations



LEON3 CPU support in Renode

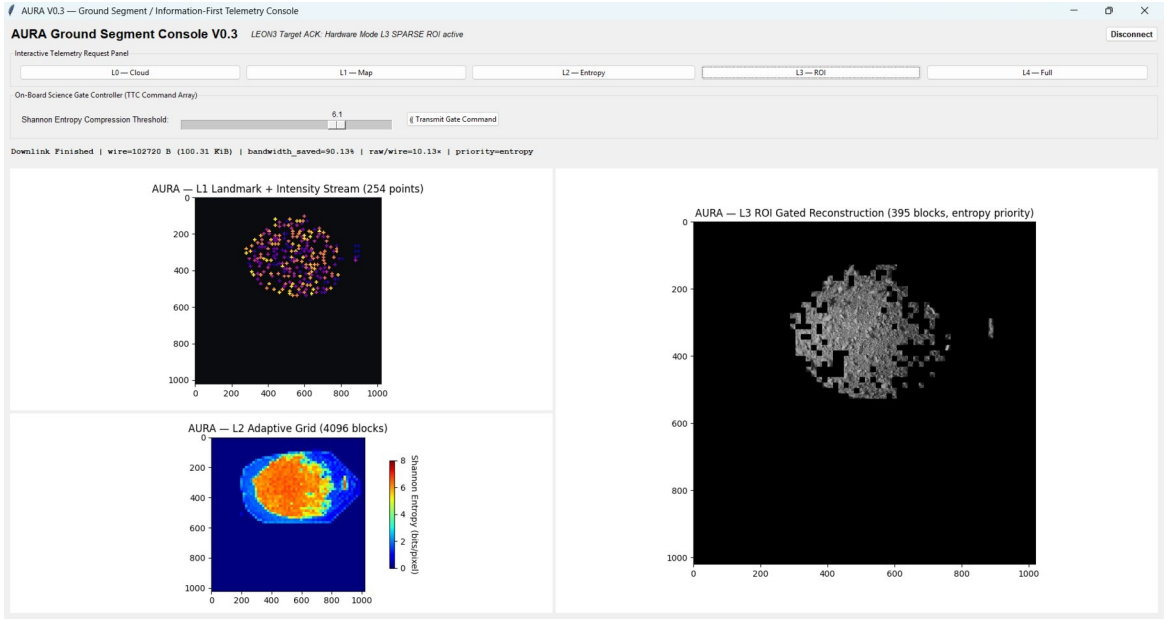
Development Road map(up to TRL7)

Today TRL4

Concept → Software prototype
→ LEON3 emulation

Next Steps

- Hardware validation
- Radiation-tolerant platforms
- Integration testing
- Mission-relevant datasets



Research Branches

- Progressive telemetry
- Information filtering
- Visual navigation
- Joint entropy sensor fusion

Long-Term Goal

TRL progression toward operational space systems.

Why a PhD?

Scientific Questions

Can entropy-based representations:

- **Can information-theoretic representations improve autonomous spacecraft operations?**
- **Reduce bandwidth requirements?**
- **Improve onboard decision making?**
- **Support visual navigation?**
- **Enable multi-sensor information fusion?**

Research Potential

AURA currently opens several research branches:

1. Data prioritization
2. Information compression
3. Visual navigation
4. Joint entropy across sensors

Beyond Technology

Personal Goal

Build a long-term academic and professional future in Germany.

Motivation

- Contribute to German research
- Participate in European space projects
- Collaborate with universities and industry



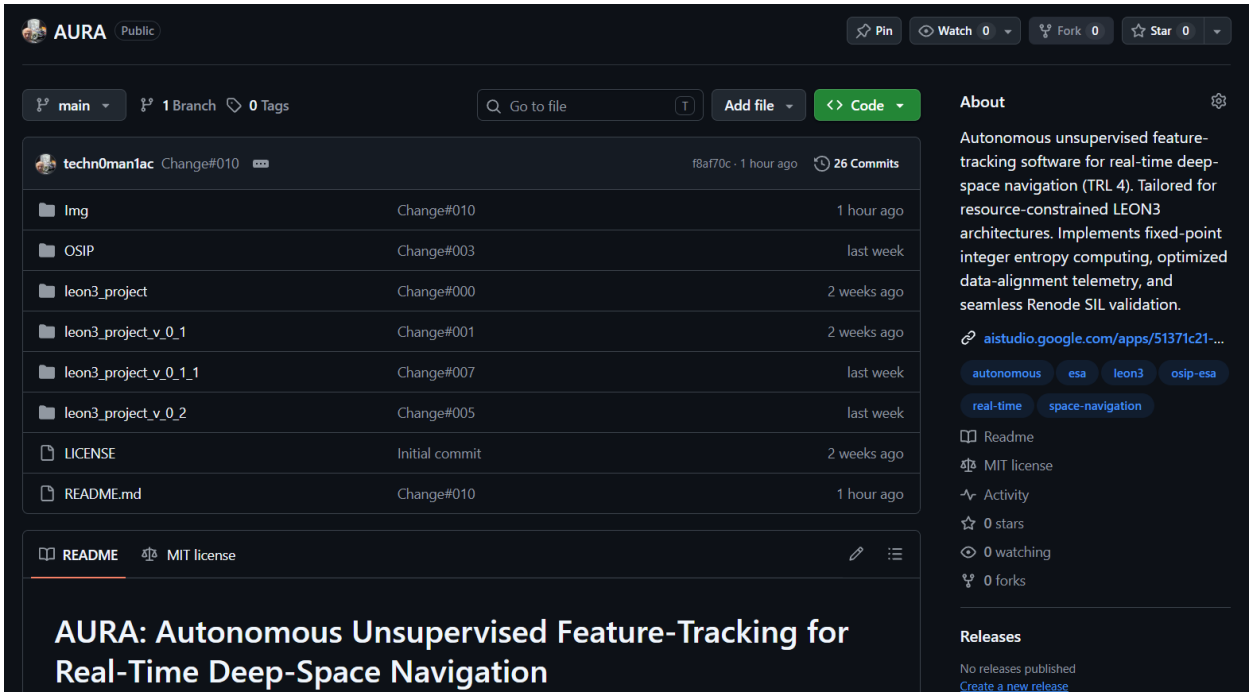
Long-Term Vision

> Growing together with the German scientific and engineering community.

AURA

Current Status

- Working prototype
- Clear research direction
- Open-source development(MIT license)
- Potential space applications



https://github.com/techn0man1ac/AURA/

Key Message

AURA explores how information itself can become a resource for autonomous space systems.

Thank You